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TO WHOM IT MAY CONCERN

This is to certify that the reported work entitled “Physics-Informed BDE Prediction via 3D Spherical Quantum Mechanics” submitted by Mr. Pratyush Upadhyay (Roll No. 230122039), currently pursuing his fourth year of undergraduate studies at the Indian Institute of Technology Guwahati, for the summer internship at my laboratory - 2026, Department of Chemistry, Indian Institute of Technology Guwahati, is based on the work carried out under my direct supervision and guidance. The project involved building an end-to-end, fully differentiable pipeline to predict Bond Dissociation Energy (BDE) and extract spatial molecular geometry. The primary components and modules utilized in this architecture include:

GINN Encoder, Morse Potential for finding radial function, Simple Harmonics for angular function, Schrodinger Wave Equation, BDE & Geometry Prediction Head. The future plan of this project includes modeling different aspects of a molecule by defining the flux of Electric Field over the derived probability density which will include uses of Faraday's Law and Gauss Law. He is a sincere, dedicated and hardworking student.

I wish her all the best for her career and future endeavours.

Place: IIT Guwahati

Prof. Parameswar K. Iyer

Date: 12/08/2026

Research Supervisor

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